

Jiasheng Huang

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EDUCATION

Santa Clara University

Santa Clara, CA

Bachelor of Science in Computer Science and Engineering

Sep 2023 – Jun 2027 (expected)

Relevant Coursework: Advanced Data Structures, Theory of Algorithms, Operating Systems, Software Engineering, Machine Learning and Data Mining, Artificial Intelligence, Web Search and Information Retrieval

EXPERIENCE

Software Engineering Intern

Milpitas, CA

Vigiles Robotics Inc. — Semantic Mapping for an LLM-Driven Inspection Robot

Jul 2026 – Present

- Developing the semantic mapping module for an LLM-driven quadruped inspection robot operating in indoor environments (offices, factories, campuses), targeting on-board deployment on Jetson AGX Orin 64GB (Ubuntu 22.04, ROS 2 Humble)
- Building a multimodal perception pipeline fusing Livox Mid-360 LiDAR (FAST-LIO2 odometry) with pan-tilt RGB-D camera streams; calibrated camera–LiDAR extrinsics via multi-pose joint ICP on real robot data, achieving 8.8 cm RMS residual
- Deploying open-vocabulary object detection with a quantized 3B-parameter VLM (ggml, CPU-only) paired with Qwen3-1.7B–generated scene-specific vocabularies (~15–17 classes per environment), producing semantically annotated maps in the osmAG format
- Designing a layered tool-calling interface that grounds the robot's LLM agent in physical space: 9 low-level map-query tools dispatched through a custom lightweight ROS 2 service RPC, wrapped by 2 LLM-facing skills routing natural-language queries (“where is X”, “how many Y”) to spatial lookup and inventory functions
- Architecting a pluggable SLAM adapter abstraction supporting FAST-LIO2 in production with reserved interfaces for ICP-class alternatives

PROJECTS

Course Planning Agent | Python, React, Gemini API, Streamlit

2025 – Present

- Built a multi-agent course planning system that parses SCU Academic Progress exports to identify degree-requirement gaps and generate personalized next-quarter schedules
- Implemented a Gemini-powered planning agent that converts natural-language preferences into course recommendations; enriched results with RateMyProfessor ratings via parallel thread execution

SecReviewer — AdaL Track 1st Place Winner | AdaL, GitHub Actions, Vite, OAuth

2026

- Built an AI-powered security review system on SylphAI's AdaL coding agent, shipped as three layers: a local pre-push git hook, a GitHub Action, and a unified review dashboard
- Engineered a pre-push hook piping git diffs into AdaL headless mode to block SQL injection, hardcoded secrets, broken auth, and unsafe deserialization before code reaches GitHub
- Proposed the “AdaL Expert Pack” framework — a reusable architecture for shipping vertical reviewers (security, performance, style) on one AI agent engine

Fitness Tracker | React, Node.js, Express, PostgreSQL, JWT

2025 – Present

- Built and deployed a full-stack nutrition tracking app (Vercel/Render/Neon), cutting personal meal-logging time by ~50% with JWT auth, REST APIs, and relational schemas

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL (PostgreSQL), JavaScript, HTML/CSS

Machine Learning: Multimodal Sensor Fusion, Vision-Language Models, LLM Agent Interfaces, RAG, LangChain, PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn

Frameworks & Tools: ROS 2, React, Node.js, Express, FastAPI, Git, Linux/Unix, CI/CD, Claude Code, Cursor

Databases & Cloud: PostgreSQL, AWS (S3, Bedrock)